

# PriceShield

## System Features and Requirements

Data Intelligence for Food Security in The Gambia

### **Document structure**

Five business-oriented system features with concise descriptions, use case flows and measurable functional requirements.

# Document overview

The document uses collapsible Word headings. Select the arrow beside a feature or requirement heading to expand or collapse its content.

| System feature                                  | Business purpose                                                                                                                             |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 1 ADP - Automated Data Processing               | Automatically imports, stores and processes new monthly food-price data without manual Excel work.                                           |
| 2 RCD - Reliable and Comparable Data            | Validates and standardises food-price data so that commodities, units and currencies can be compared reliably.                               |
| 3 MON - Price Monitoring and Country Comparison | Shows national and regional price trends and compares matching commodities between The Gambia and Guinea-Bissau.                             |
| 4 FEW - Forecasting and Early Warning           | Forecasts future commodity prices and warns about significant expected price increases.                                                      |
| 5 REP - Monitoring and Reporting                | Provides authorised analysts with a secure Power BI report for fast meeting analysis and controlled reporting to government decision-makers. |

# 1 Automated Data Processing

## Description

Automatically imports, stores and processes new monthly food-price data without manual Excel work.

## Use Case Flow

1. A new monthly food-price file becomes available.
2. PriceShield retrieves and stores the original file.
3. The system keeps the records for The Gambia and Guinea-Bissau.
4. The system starts validation and transformation.
5. If processing succeeds, the system publishes the new data.
6. If processing fails, the system keeps the previous valid dataset and informs Fatou.

## Functional Requirements

### REQ-ADP-01 Automated ingestion

The system shall automatically ingest every new monthly food-price dataset. Compliance is achieved when all valid monthly deliveries are processed without manual file upload.

### REQ-ADP-02 Raw-data preservation

The system shall store every received source file unchanged before processing. Compliance is achieved when each file has a unique file ID, ingestion timestamp and source identifier.

### REQ-ADP-03 Country filtering

The system shall retain only observations for The Gambia and Guinea-Bissau in the curated analytical dataset. Compliance is achieved when the curated dataset contains no records from other countries.

### REQ-ADP-04 Processing deadline

The system shall complete the monthly processing before the updated dashboard is published. Compliance is achieved when at least 95% of valid monthly deliveries are published within 24 hours of arrival.

### REQ-ADP-05 Failed-load protection

The system shall preserve the latest validated dataset when a processing run fails. Compliance is achieved when no failed delivery replaces the current production data.

## 2 Reliable and Comparable Data

### Description

**Validates and standardises food-price data so that commodities, units and currencies can be compared reliably.**

### Use Case Flow

1. The system receives the newly ingested source data.
2. It checks mandatory fields, prices and dates.
3. It identifies missing months and duplicate observations.
4. It standardises commodity names and units.
5. It converts comparable prices into USD.
6. Valid records enter the curated dataset.
7. Invalid or incompatible records enter quarantine with an explanation.

### Functional Requirements

#### REQ-RCD-01 Data validation

The system shall validate country, region, market, commodity, price type, currency, unit, month, year and price on data type and possible ranges before publication. Compliance is achieved when 100% of published records pass all mandatory validation rules and alerts are raised for unpassed records.

#### REQ-RCD-02 Missing and invalid data

The system shall distinguish missing monthly observations from invalid price values. Compliance is achieved when missing months are identified, non-positive or non-numeric prices are quarantined, and no missing observation is interpreted as zero.

#### REQ-RCD-03 Duplicate prevention

The system shall identify duplicates using country, region, market, commodity, price type, unit, month and year as the business key. Compliance is achieved when the published dataset contains zero duplicate business keys.

#### REQ-RCD-04 Unit standardisation

The system shall convert commodity prices into a standard unit only when an approved conversion factor is available. Compliance is achieved when every converted observation retains its original unit, standard unit and conversion factor.

#### REQ-RCD-05 Currency standardisation

The system shall convert comparable prices from Gambian dalasi and West African CFA francs into United States dollars using an approved monthly exchange rate. Compliance is achieved when each converted value retains its original price, original currency, applied exchange rate, exchange-rate date and converted USD value.

#### REQ-RCD-06 Invalid comparison handling

The system shall exclude records that cannot be standardised from direct comparisons. Compliance is achieved when every excluded record remains available with a documented exclusion reason.

## 3 Price Monitoring and Country Comparison

### Description

Shows national and regional price trends and compares matching commodities between The Gambia and Guinea-Bissau.

### Use Case Flow

1. Fatou selects a commodity and reporting period.
2. The dashboard displays its national price trend.
3. Fatou selects a region or market for further detail.
4. She activates the comparison with Guinea-Bissau.
5. The system checks commodity, unit, price type and period compatibility.
6. It displays the price difference or explains why no valid comparison is possible.

### Functional Requirements

#### REQ-MON-01 National price trends

The system shall display monthly national price trends for every available commodity. Compliance is achieved when users can select a commodity and view all valid national observations for the selected period.

#### REQ-MON-02 Regional drill-down

The system shall allow users to drill down from national results to available regional and market observations. Compliance is achieved when selecting a region or market updates the analysis without changing the selected commodity and period.

#### REQ-MON-03 Comparable observations

The system shall compare prices only when observations represent the same canonical commodity, price type, standard unit and month. Compliance is achieved when 100% of published comparisons meet all four conditions.

#### REQ-MON-04 Relative price changes

The system shall calculate month-on-month and year-on-year percentage changes for every valid commodity series. Compliance is achieved when both measures are displayed whenever the necessary reference observations exist.

#### REQ-MON-05 Country price difference

The system shall calculate the relative price difference between comparable commodities in The Gambia and Guinea-Bissau. Compliance is achieved when every comparison identifies the reference country and uses comparable USD prices from the same month.

#### REQ-MON-06 Data coverage

The system shall display the data coverage of each selected commodity series. Compliance is achieved when users can see the number of available observations, percentage of missing months and latest observation date.

# 4 Forecasting and Early Warning

## Description

Forecasts future commodity prices and warns Fatou about significant expected price increases.

## Use Case Flow

1. New validated monthly prices become available.
2. The system checks whether each commodity has sufficient historical data.
3. It updates the 12-month forecast for eligible commodities.
4. The system compares observed and forecast price changes with approved thresholds.
5. If a threshold is exceeded, it generates an alert.
6. Fatou reviews the commodity, location, expected change and forecast uncertainty.

## Functional Requirements

### REQ-FEW-01 Forecast eligibility

The system shall evaluate the forecasting eligibility of every country-commodity series and if labeled as eligible execute the forecasting process. Compliance is achieved when each series receives an eligibility status, observation count, temporal coverage rate and eligibility reason.

### REQ-FEW-02 Minimum forecast data

The system shall produce a time-series forecast only for series with at least 24 valid monthly observations and 70% temporal coverage. Compliance is achieved when ineligible series display “insufficient data” instead of an unsupported forecast.

### REQ-FEW-03 Forecast horizon

The system shall generate a 12-month price forecast for every eligible commodity series. Compliance is achieved when each eligible series contains twelve monthly point estimates.

### REQ-FEW-04 Forecast uncertainty

The system shall communicate forecast uncertainty. Compliance is achieved when every published forecast includes a prediction interval and at least one out-of-sample error metric.

### REQ-FEW-05 Relative alerts

The system shall generate an early-warning alert when a forecast price increase exceeds a commodity-specific threshold (adjustable by an authorised analyst, with a default threshold of 10%). Compliance is achieved when every alert displays the commodity, location, percentage change, comparison period, applied threshold and timestamp.

## 5 Monitoring and Reporting

### Description

Enables Fatou to analyse government actions in Power BI, answer questions during meetings and export concise reports for ministers.

### Use Case Flow

1. Validated monthly data becomes available in Power BI.
2. Fatou opens the report using her authorised organisational account.
3. She filters by commodity, country, region, market or period.
4. During a meeting, Fatou uses the report to answer specific questions.
5. She exports the selected pages or visuals as PDF or PowerPoint.
6. Ministers receive the exported information without accessing the analytical system.

### Functional Requirements

#### REQ-REP-01 Interactive analysis

The Power BI report shall allow authorised analysts to filter by commodity, country, region, market and reporting period. Compliance is achieved when changing a filter updates all connected report visuals consistently.

#### REQ-REP-02 Data freshness information

The Power BI report shall display the latest successful refresh time and latest available observation month. Compliance is achieved when both values are visible in the report and included in exported report pages.

#### REQ-REP-03 Controlled report export

The Power BI report shall allow authorised analysts to export the data of the presented visuals. Compliance is achieved when the export reflects the active filters generation timestamp.

#### REQ-REP-04 Report performance

The Power BI report shall load pages and update visuals within five seconds for at least 95% of interactions. After extended inactivity, the first request may take up to 30 seconds due to server reactivation.

#### REQ-REP-05 Authenticated report access

The system shall restrict Power BI access to authorised organisational accounts and block anonymous or public access, so that internal government information remains protected.